

SMART LAND REPURPOSING: A DATA-DRIVEN FRAMEWORK FOR SUSTAINABLE USE OF BARREN AND DEGRADED LANDS

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1 INTRODUCTION AND PROBLEM STATEMENT

Less than 2% of the U.S. wildfire budget is spent on recovery, despite most wildfire-related costs and damages occurring after the initial fire event (Crowley et al., 2023). Recovery decisions must be made within days (USDA Forest Service, 2025) and shape long-term outcomes, yet current workflows rely on manual field surveys and fragmented image analysis. These methods are slow, expose ground teams to hazardous post-fire environments, and limit the effectiveness of interventions, especially under tight resource constraints (U.S. Geological Survey, 2018; Western Fire Chiefs Association, 2024; Schlunegger et al., 2023).

There is an urgent need for automation and data integration to optimize recovery planning. AI is uniquely positioned to address this gap: machine learning models can rapidly analyze post-fire satellite imagery, assess ecological risk, and generate transparent, data-driven recovery recommendations aligned to user priorities (National Center for Disaster Preparedness, 2025). By reducing field time and standardizing analysis, AI-based tools can improve the speed, safety, and consistency of decisions, ultimately enabling more effective and accountable landscape recovery.

Problem Statement: Existing manual and fragmented post-fire recovery assessments are inadequate for making timely, safe, and effective recovery decisions. *TerraNova* is proposed as an interactive AI-powered platform that aims to:

- Reduce assessment time and field risks for experts
- Improve the precision and transparency of recovery decisions
- Make actionable recovery guidance accessible to both professionals and community members.

2 INTERFACE DESIGN

2.1 DESIGN JUSTIFICATION

TerraNova uses a map-centered interface because post-wildfire recovery decisions are spatial. Users must locate a burn footprint, compare conditions across locations, and prioritize actions at the parcel or block level. A map-first layout reduces context switching by letting users interpret results directly on the landscape and drill down only when needed.

We designed the interface around three goals. First, it should support quick situation awareness, so a user can identify high-severity and high-risk zones within a few interactions. Second, it should enable defensible planning by making model outputs comparable rather than hiding them behind a single combined score. Third, it should support handoff and coordination, since recovery planning requires sharing results with field teams and stakeholders.

These goals drove the core layout decisions:

- A split layout where the map remains the primary workspace and controls live in a side panel
- Layer-based visualization so burn severity, reburn risk, and recommended next steps can be viewed as distinct overlays

- Progressive disclosure so the default experience remains simple while advanced analysis remains accessible
- Report-style outputs and summary dashboards to produce a shareable artifact, not only an interactive view.

We also structured the interaction to support two common user needs without forcing one workflow. A planning experience emphasizes comparison across layers, prioritization, and export. A risk-check experience emphasizes a simpler summary view and quick interpretation for non-expert decisions.

2.2 CHANGES MADE AND RATIONALE:

Early iterations of the interface followed a linear pipeline pattern: a user provided input, the system produced a result, and the UI mainly displayed outputs. This structure was useful for demonstrating the modeling pipeline, but it created three gaps for decision support. It did not support iterative exploration, it did not make it easy to compare different model signals on the same area, and it did not match how recovery planning typically happens in practice, where users revisit assumptions and refine priorities.

Later iterations moved the interface toward an interactive decision workflow. We added search and filtering so users can start from a real fire event rather than treating the system as a generic file uploader. We reorganized the main experience around selecting a fire, zooming to its footprint, and exploring model outputs as map layers. This change was necessary because recommendations must be tied to a specific place, and visual comparison across severity, risk, and actions is central to prioritization.

We also introduced features that support trust and communication. An explanation-oriented panel enables contextual justification tied to what is visible on the map. A dashboard view summarizes severity distribution and recommended mitigation mix in a compact format, and report export supports coordination with teams. These changes were necessary because decision support requires explanation and a shareable summary that fits into existing planning workflows.

Finally, we refined usability tradeoffs under time pressure by supporting adjustable granularity. Users can begin with a fast overview and switch to more detailed views when needed, which keeps the interface responsive while still enabling deeper analysis.

Appendix A.2 shows the final screens that reflect these changes, including the map layers, dashboard, and report export.

3 MACHINE LEARNING BACKEND

TerraNova’s backend is implemented as a three-stage pipeline: (1) pixel-level burn severity segmentation from satellite imagery, (2) parcel-level reburn risk prediction using linked climate and hydrologic covariates, and (3) multi-objective optimization that assigns “best next steps” to each parcel.

3.1 STAGE 1: BURN SEVERITY SEGMENTATION (U-NET)

We use the Monitoring Trends in Burn Severity (MTBS) dataset as our primary source of labeled post-fire imagery. For each MTBS fire event, we extract the post-fire Landsat scene and the corresponding burn-severity mask (dNBR6), and crop them into fixed-size patches centered on the burn footprint. Each patch contains six spectral bands (Blue, Green, Red, NIR, SWIR1, SWIR2), which are stacked into a 6-channel input tensor for the model. Cropping around the fire perimeter rather than uniform tiling reduces the dominance of background “unburned” pixels and ensures that most training patches contain meaningful burned area.

We train a U-Net-style encoder-decoder architecture that outputs pixel-wise probabilities over the standard MTBS burn-severity classes (e.g., Unburned, Low, Moderate, High, Increased Greenness, Non-Processing). The encoder consists of four convolutional blocks with max-pooling; the decoder mirrors this with upsampling and skip connections. To better handle class imbalance, we use a loss

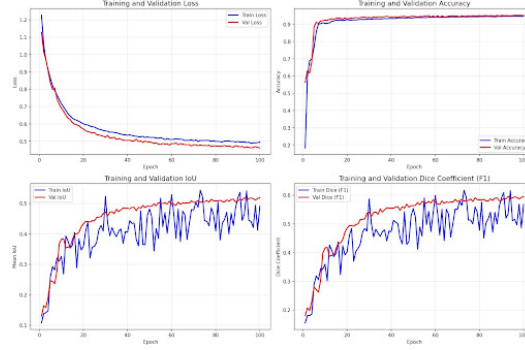


Figure 1: Training dynamics of the U-Net model: training and validation loss, accuracy, IoU, and Dice coefficient across epochs.

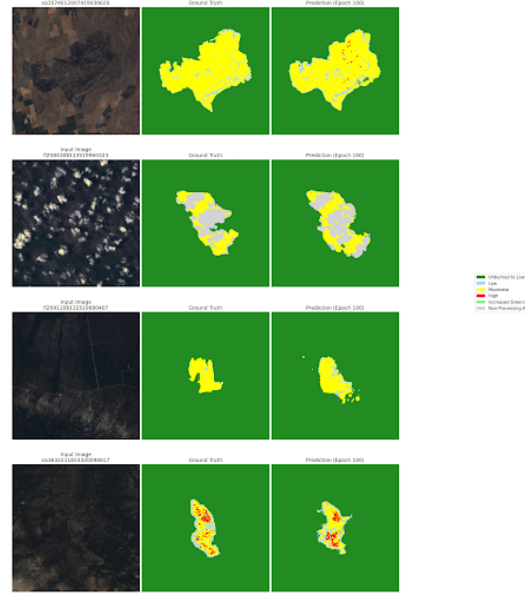


Figure 2: Qualitative U-Net burn-severity predictions on held-out MTBS fires. For each event we show the post-fire RGB image (left), MTBS ground-truth burn-severity mask (middle), and Terra-Nova prediction at Epoch 100 (right).

that combines Dice loss and cross-entropy, and apply standard data augmentation (random rotations, flips, and contrast changes). The resulting severity maps serve as the primary spatial input to later stages and are also directly visualized in the user interface as burn-severity layers.

3.2 STAGE 2: PARCEL-LEVEL REBURN RISK PREDICTION (RANDOM FOREST)

To estimate how likely a given parcel is to burn again, we construct a tabular dataset by joining MTBS-derived information with climate and hydrologic covariates at matched locations. For each parcel (coarse grid cell or management unit), we aggregate:

- Burn-severity statistics derived from the U-Net predictions.
- Burn-history features from MTBS, including how many distinct MTBS fires have affected that parcel.
- Environmental covariates from USGS/NOAA sources (Section 4): elevation, long-term average temperature and precipitation, soil moisture, groundwater depth, and soil pH.

We define a binary reburn label `reburn` that indicates whether the parcel experienced at least one additional fire within the MTBS time window. A Random Forest classifier then maps parcel features x to a reburn probability

$$p = P(\text{reburn} = 1 \mid x),$$

which we interpret as a continuous reburn-risk score in $[0, 1]$.

3.3 STAGE 3: MULTI-OBJECTIVE LAND-USE OPTIMIZATION (GENETIC ALGORITHM)

The final stage translates predicted risk and environmental context into concrete management actions. We model the landscape as a set of parcels, each of which can be assigned one of four high-level actions: *Protect unburned refugia*, *Targeted erosion control*, *Replant high severity*, *Fuel breaks and buffer* or *Monitor and wait*. Each action carries a cost and an expected effect on risk reduction, soil stability, water resources, and long-term ecological resilience.

We use a genetic algorithm to search over possible parcel–action assignments. Each candidate solution encodes the action chosen for every parcel, and is scored using a scalar objective that combines the different criteria:

$$F = 0.35(1 - R) + 0.15S + 0.20W + 0.20L - 0.10C,$$

where R is the reburn-risk score from Stage 2, S is a soil-stability score, W reflects water-related impacts, L approximates long-term ecological resilience, and C is the normalized intervention cost. Through iterative selection, crossover, and mutation, the algorithm converges toward spatially coherent action layouts that balance risk reduction, ecological suitability, and budget. These optimized recommendations are then surfaced in the TerraNova interface as parcel-level “next best steps.”

3.4 ADAPTATIONS, CHALLENGES, AND RATIONALE

Burn Severity Segmentation Initially, the U-Net segmentation model yielded poor performance distinguishing high-severity burn pixels, largely due to extreme class imbalance and the dominance of unburned/low pixels in the MTBS data. Validation metrics such as Dice and IoU were lower than anticipated, and qualitative review found the model failed to delineate higher severity zones. To address this, we changed our patch selection strategy to aggressively crop training patches around burn footprints, improving the representation of burnt regions and raising our Dice scores substantially. We also tuned the loss function to better account for small, high-severity classes, using a weighted combination of Dice and cross-entropy loss.

Reburn Risk Prediction Reburn risk prediction was introduced after the first iteration, following feedback from JHU faculty and agency experts who requested actionable risk assessments beyond burn severity mapping. This required integrating tabular parcel-level features from MTBS, NOAA, and USGS datasets and training a random forest classifier on temporal fire histories. This addition was a direct response to user evaluations and shifted our system focus beyond descriptive burn maps to predictive risk guidance for future events.

Data Selection for Model Training The original proposal planned to train the segmentation model across all US state MTBS fires, but this proved infeasible due to cost and resource constraints associated with preparing, storing, and processing gigabytes of satellite fire bundles. Instead, we limited training to Florida and Alabama fires and reserved California fires for held-out testing. This allowed efficient model development while still supporting geographically diverse evaluation.

4 DATA

Our system integrates remote sensing imagery from MTBS with climate and hydrologic covariates from USGS/NOAA to support both pixel-level burn mapping and parcel-level risk modeling.

4.1 REMOTE SENSING AND BURN-SEVERITY LABELS (MTBS)

We rely on the Monitoring Trends in Burn Severity (MTBS) archive, which provides fire-event–level ZIP bundles containing pre- and post-fire Landsat imagery, burn-severity masks (dNBR6/NBR6), and metadata for large wildfires across the United States. For each selected fire event, we:

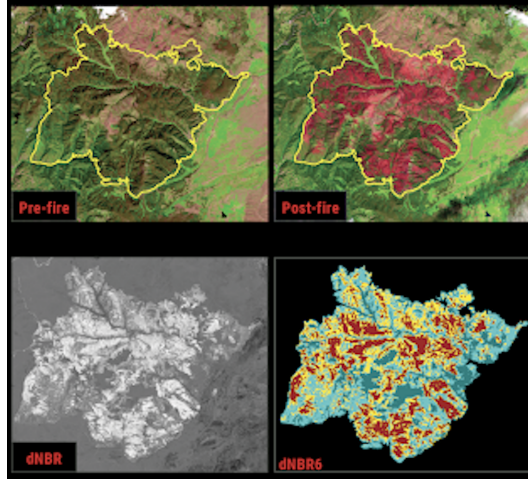


Figure 3: Overview of MTBS used in TerraNova, including pre- and post-fire Landsat imagery, continuous burn severity indices (dNBR, RdNBR), and the 6-class thematic burn-severity layer that we use as ground-truth labels.

1. Download and unpack the MTBS event ZIP file.
2. Extract the post-fire surface reflectance image (Landsat 5/7/8), keeping six spectral bands: Blue, Green, Red, NIR, SWIR1, and SWIR2.
3. Pair the post-fire image with the corresponding burn-severity mask (dNBR6).
4. Parse the XML metadata to obtain fire name, year, acquisition date, sensor, spatial extent (west/east/north/south bounding coordinates), and, when available, fire size (acres).

We then crop each post-fire image and its mask into fixed-size patches centered on the burn perimeter, discarding tiles with negligible burned area. This yields N_{patch} curated (image, mask) pairs at 512×512 resolution after filtering (e.g., removing tiles with negligible burned area and invalid/no-data regions). A master CSV links each patch back to its originating event and files, with columns such as `fire_id`, `state`, `pre_fire_file`, `post_fire_file`, `mask`, `xml_file`, `fire_name`, `fire_year`, `date`, and `post_fire_sensor`. This CSV is the entry point for the segmentation stage and for aligning imagery with external covariates.

Dataset statistics: The MTBS burn severity dataset used in this project comprises 5,773 image/mask patches for model training and validation, sourced from fire events in Florida and Alabama, and 2,078 held-out test samples from multiple California fires. Each patch is 512×512 pixels and contains six Landsat spectral bands. The dataset spans fire events from 1984 to 2024 and is split by fire event to ensure robust generalization, with California reserved entirely for testing. In total, the class distribution across all labeled pixels is: Unburned (62%), Low severity (18%), Low-Moderate (7%), Moderate (7%), High (5%), and Increased Greenness (1%). This sample size and diverse representation surpass typical remote sensing segmentation studies, providing sufficient coverage for effective training and statistically sound evaluation.

4.2 CLIMATE AND HYDROLOGIC COVARIATES (USGS/NOAA)

To capture drivers of reburn risk beyond immediate burn severity, we enrich the MTBS-derived dataset with parcel-level environmental variables assembled from USGS and NOAA products. For each parcel, we extract or aggregate the following features:

- **Topography and soils (USGS):** elevation in meters (`elevation_m`), volumetric soil moisture percentage (`soil_moisture_pct`), groundwater depth in feet (`gw_depth_ft`), and soil pH (`ph_val`).

- **Climate (USGS/NOAA composites):** long-term average air temperature in degrees Celsius (`avg_temp_c`), annual precipitation in millimeters (`annual_precip_mm`), and event-period precipitation (`precip_mm`).

These variables are sampled or averaged at the same locations as the MTBS fire pixels, and then aggregated to the parcel level (e.g., mean or median over all pixels within a parcel polygon). In combination with burn-severity statistics, they provide a compact representation of the physical environment relevant to post-fire recovery and reburn dynamics.

Dataset statistics: The constructed dataset contains $N=500$ fire-event records (500 unique `fire_ids`) spanning 1984–2024 (41 unique fire years), all in California. The reburn label is available for all 500 events with a positive rate of $p(y=1) = 0.618$ (309 positive, 191 negative). Covariate ranges (min–max; mean) are: elevation 57.7–2846.7 m (894.9), average temperature 4.32–28.81 °C (19.23), precipitation 228.7–1423.9 mm (657.9), soil moisture 2.43–27.51% (12.90), groundwater depth 5.0–178.28 ft (65.66), and soil pH 5.89–8.40 (7.23). Bounding-box coordinates are present for 491/500 events (98.2%); land-burned frequency has 9/500 missing values (1.8%).

4.3 REBURN LABELS FOR RISK MODELING

For reburn risk prediction, we construct a binary label per parcel that captures whether the parcel experienced at least one subsequent fire after an index fire. We derive labels by intersecting parcel polygons with historical fire perimeters from our fire-event inventory and marking a parcel as positive if it overlaps any later fire footprint within the observation window.

Label construction: For each index fire event (year y_0), we consider all recorded fire perimeters in years $(y_0, y_0 + \Delta]$ and compute parcel–perimeter overlap. A parcel is labeled $y=1$ if its overlap with any later fire exceeds a minimum threshold τ (e.g., non-zero overlap or at least τ percent of parcel area), and $y=0$ otherwise. We exclude parcels with incomplete temporal coverage (e.g., missing covariates or uncertain event boundaries) when label assignment is ambiguous.

Dataset statistics: The resulting dataset contains N_{examples} labeled parcels across years $[Y_{\min}^{\text{reburn}}, Y_{\max}^{\text{reburn}}]$ with a positive rate of $p(y=1)$. We report the distribution of time-to-reburn for positive parcels (mean/median years) as an interpretability check, since the binary label collapses timing into a single indicator.

4.4 ADVANTAGES, DISADVANTAGES, AND LIMITATIONS

Advantages:

- **Reproducible public sources.** The data stack relies on well-documented, publicly available datasets, which supports reproducibility and easier validation of results.
- **Paired imagery and labels for supervised learning.** MTBS event bundles provide paired Landsat imagery and burn-severity products with event metadata, enabling supervised burn-severity segmentation and consistent linkage of patches to their originating fire events.
- **Multispectral signal for burn characterization.** Using Landsat spectral bands (Blue, Green, Red, NIR, SWIR1, SWIR2) improves sensitivity to vegetation change and burn signals compared to RGB-only imagery.
- **End-to-end alignment across stages.** The master CSV connects each patch to its fire-level metadata and files, which simplifies traceability across segmentation, risk modeling, and downstream analysis.
- **Richer context for risk modeling.** Adding climate, topographic, and hydrologic covariates complements burn-severity summaries and provides additional explanatory signal for reburn risk beyond immediate damage indicators.

Disadvantages, limitations, and potential bias:

- **Mapping and Alignment Challenges.** Environmental layers are sampled at different spatial resolutions and sometimes have coarse or mismatched spatial grids compared to MTBS fire perimeters and satellite imagery. Latitude/longitude mapping between fires and USGS parcels required approximations and at times led to positional uncertainty that might affect parcel-level risk modeling.
- **Generalization Limits.** The model was trained and tested on data from Florida, Alabama, and California, which may introduce bias. The system’s performance in other regions with different fire characteristics is uncertain, for both segmentation and risk prediction.
- **Temporal mismatch.** Climate and hydrologic variables are often long-term averages or annual measures, which may not fully correspond to conditions during each specific fire event, reducing accuracy of incident-specific predictions.
- **Label uncertainty from index-based severity.** MTBS burn-severity classes are derived from spectral indices and thresholding, so they provide an operational label rather than a direct physical measurement. Ambiguity is highest near class boundaries, which can cap segmentation accuracy and introduce noise into parcel-level severity summaries.
- **Event selection bias (MTBS coverage).** MTBS burn severity masks are based on remote-sensing indices and expert thresholds, so boundaries between severity classes can be ambiguous. Pixel-level noise can affect the model’s predictions and downstream parcel summaries.
- **Sensor and acquisition bias (domain shift).** Events use different Landsat sensors (5/7/8) and post-fire acquisition dates vary. Differences in sensor characteristics and timing (including smoke, haze, and phenology) can systematically shift spectral signatures and bias predictions toward conditions overrepresented in training.
- **Spatial and temporal resolution constraints.** Landsat inputs are medium resolution with multi-day revisit cycles, which limits capture of fine-grained heterogeneity and rapidly changing post-fire conditions. Parcel-level outputs can therefore be less reliable for small features that fall below the effective resolution of the imagery.
- **Reburn label simplification.** The reburn target is a simply indicates whether a parcel burned again within the MTBS time window so it does not capture recurrence timing.
- **Downstream decision limits from incomplete constraints.** Our optimization scores are based on estimated costs and ecological effects; they do not fully account for legal, cultural, or stakeholder constraints. As a result, recommendations are best used as decision support rather than strict action plans.

EVALUATION

5.1 EVALUATION OF THE ML BACKEND

We conducted a comprehensive automated evaluation of our ML backend using standard metrics and test splits across its major components: segmentation, reburn risk prediction, genetic planning, and system performance.

5.1.1 SEMANTIC SEGMENTATION METRICS

Segmentation performance was measured by pixelwise accuracy and overlap-based metrics. The model achieved an overall test accuracy of 0.95. The model achieved a mean Dice coefficient of approximately 0.70, indicating strong region-level agreement between predicted and ground-truth burn masks. The corresponding mean Intersection-over-Union (IoU) was approximately 0.55, which is expected given the stricter overlap criterion of IoU. Per-class IoU and Dice scores highlight strong segmentation of the background class (IoU: 0.96, Dice: 0.98), with variable results for more challenging foreground classes.

5.1.2 REBURN RISK PREDICTION

For binary classification of reburn risk, a Random Forest was tested on a dataset of 500 fire events (310 reburn / 190 no-reburn), using a grouped split (train: 350, val: 75, test: 75). On the test set,

the model achieved a ROC-AUC of 0.9912, F1 score of 0.9888, and accuracy of 0.9867. Precision was 0.9778 and recall was 1.0000, meaning the model identified all reburn events (no missed re-burns) and made very few false reburn predictions. The most influential feature was Land Burn Frequency (importance: 0.8195), suggesting that historical fire frequency is the strongest predictor of reburn risk in our dataset.

5.1.3 BEST-NEXT-STEPS PLANNER (GENETIC ALGORITHM)

Planning performance was assessed using a multi-objective score aggregating risk, cost, and ecological value (average score ≈ 0.98 on evaluation fires). With the optimal Genetic Algorithm (GA) configuration (population = 200, generations = 150), the GA achieved a mean objective improvement of +0.025 over the greedy baseline, demonstrating effective optimization of the multi-faceted objective. This improvement was achieved with a mean runtime of 3.46 s (95th percentile: 15.05 s).

5.1.4 SYSTEM PERFORMANCE & RELIABILITY

API latency and reliability were assessed for both segmentation and planning endpoints. For the `/api/segment` endpoint ($N = 25$), the median latency (p50) was 529.85 ms and the mean was 547.80 ms, with a 100% success rate and no errors. For the `/api/ask` endpoint ($N = 10$), the median latency was 713.78 ms, with a 100% success rate and no errors.

5.2 QUALITATIVE ANALYSIS AND SYSTEM EVALUATION: SOCIOTECHNICAL EFFICACY OF TERRANOVA

We conducted a qualitative evaluation of TerraNova by focusing on its utility, usability, and trustworthiness in real-world post-wildfire recovery workflows. Our mixed-methods approach included scenario-based usability testing and semi-structured interviews with a stratified sample of eight participants, enabling an insight into the system’s sociotechnical fit. A detailed summary table of qualitative findings, participant reactions, and design implications is provided in ??.

5.2.1 USER COHORTS AND EVALUATION PROTOCOL

Participants were purposively sampled to achieve high information power. The sample ($N=8$) encompassed:

Group	n	Profile	Evaluation Focus
Environmental Experts	2	Faculty (Ecosystem Recovery)	Model ecological validity; transparency; uncertainty
Technical Experts	2	GIS/Computer Vision Specialists	Workflow integration; latency; data export
General/Lay Users	4	Community Members/Stakeholders	Clarity of risk scores; map and AI interpretability

Table 1: Qualitative study participant groups and primary evaluation foci.

The study protocol mirrored a realistic disaster response workflow: participants explored a historical wildfire case with the TerraNova platform, moving sequentially through assessment (burn severity), prediction (reburn risk), prescription (GA-based recommendations), and open-ended interrogation (Ask AI), before debriefing interviews.

5.2.2 KEY THEMES AND INSIGHTS

Screen recordings, think-alouds, and post-task interviews were inductively coded and mapped to sociotechnical acceptance constructs (usefulness, ease of use, trustworthiness), as summarized below:

5.2.3 STRENGTHS: FACILITATORS OF EFFECTIVE RECOVERY OPERATIONS

- **Workflow Integration and Shared Situational Awareness:** By aggregating geospatial data, severity/risk prediction, and action recommendations in a single dashboard, Terra-

Nova addressed the primary “fragmentation” pain point in post-fire assessment. Both technical and general users noted decreased cognitive load and faster pathway from data acquisition to actionable insight.

- **Democratization of Complex Analysis:** Abstracting remote sensing indices and risk metrics behind intuitive categories and a natural language interface broadened accessibility and community engagement.
- **Scenario Comparison for Pattern Recognition:** Support for rapidly contrasting past and present fire cases enabled users (especially domain experts) to apply recognition-primed decision-making strategies.
- **Engineering Modularity:** Decoupled model architecture (U-Net, Random Forest, GA planner) allows future extension/replacement with minimal technical friction, critical in fast-evolving wildfire science.

5.2.4 DEFICIENCIES: BARRIERS TO ADOPTION AND TRUST

- **No Uncertainty or Confidence Display:** Both risk predictions and AI explanations lacked confidence bounds, hampering operational trust and appropriate calibration of response aggressiveness.
- **Limited Contextualization of Recommendations:** Action plans do not account for real-world constraints (e.g., budget, land ownership, seasonality) due to unavailable or fragmented data, which was beyond the scope of a one-semester project. Some predictions may therefore be infeasible or misaligned with operational realities.

5.2.5 USER-DRIVEN DESIGN RECOMMENDATIONS

Emerging from participant feedback are several actionable improvements:

- **Interactive Constraint/Weight Editor:** Enable users to dynamically adjust cost function weights (e.g., prioritizing ecological value over cost) and immediately visualize impacts on recommendations.
- **Uncertainty Visualization:** Integrate confidence intervals or UI affordances (e.g., fuzzy boundary maps, confidence sliders) to communicate prediction reliability.
- **Explicit Data Provenance:** Link all AI/LLM outputs to underlying datasets, cite sources, and clarify recency.
- **Constraint-aware Planning:** Allow operators to specify operational constraints (access, legal, financial) that modulate the optimization.
- **Temporal Transparency:** Timestamp all data layers and show “confidence decay” as time since sensing increases.

5.2.6 CONCLUSION AND IMPLICATIONS FOR FIELD DEPLOYMENT

Dimension	Current System	User-Recommended Enhancement
Transparency	Static cost functions, no AI/citation	Adjustable weights, explicit provenance/citation
Trust/Calibration	Deterministic, no uncertainty display	Confidence intervals, uncertainty-aware UI
Usability	Unified dashboard, clear maps	Dynamic scenario overlays, constraint input
Adaptability	Modular, updatable models	Human-in-the-loop planning, longitudinal data
Decision Quality	Fast, actionable recommendations	Context-aware, explainable recommendations

Table 2: System fit and user-prioritized enhancements for trustworthy, effective deployment.

Our qualitative analysis confirms that TerraNova offers transformative improvements in post-fire recovery workflow integration, data accessibility, and decision support. While future iterations should prioritize enhancing transparency, enabling interactive human control, and systematically visualizing both data provenance and model uncertainty, the promising results to date demonstrate TerraNova’s strong potential as a foundation for next-generation, user-centered disaster recovery technology.

6 IMPLICATIONS

6.1 SUPPORTING DECISION-MAKING

Our system demonstrates that combining machine learning outputs with an integrated, interactive interface can significantly improve post-fire decision-making workflows. Instead of relying on static PDF reports, fragmented datasets, or expert-only interpretation, users can move through burn severity assessment, reburn risk estimation, and recovery recommendations within a single environment. This consolidation reduces fragmentation and supports quicker sensemaking in time-sensitive post-fire contexts.

Evaluation sessions showed that the system’s primary value lies not in predicting risk alone, but in helping users translate risk into actionable next steps. Planners and land managers, in particular, benefited from seeing how different recovery actions trade off risk reduction, cost, soil stability, and water impact, rather than having to infer these relationships manually.

6.2 HUMAN-CENTERED AI, INTERPRETABILITY, AND TRUST

A key implication of our project is the importance of interpretability in AI-assisted environmental decision support. User feedback consistently emphasized the need to understand why a recommendation was generated, not just what the recommendation was. Features such as confidence indicators, feature breakdowns, and plain-language summaries helped users assess whether model outputs aligned with their expectations and domain knowledge.

This reinforces a core concept discussed in class: systems that influence real-world decisions must prioritize transparency and trust over raw predictive accuracy. Our evaluation showed that even moderately accurate models were perceived as useful when their reasoning was visible and uncertainty was communicated clearly.

6.3 USER EXPERIENCE AND COGNITIVE LOAD

The project highlighted how interface design directly shapes usability and adoption. Users valued the ability to complete end-to-end analysis without switching tools, visually compare scenarios, and explore trade-offs interactively. These design choices reduced cognitive overhead compared to traditional post-fire workflows.

At the same time, feedback revealed limitations in UX maturity. This points to the need for progressive disclosure and role-aware interfaces that adapt complexity based on user expertise. The implication is that effective decision-support systems must balance flexibility with simplicity to avoid overwhelming non-expert users.

6.4 SOCIETAL AND PUBLIC-GOOD IMPLICATIONS

Because our system operates in a post-disaster context, it carries important public-good implications. By making wildfire impacts and recovery options accessible not only to professionals but also to community members and homebuyers, the system helps reduce information asymmetry between experts and the general public. This can support more equitable and informed decisions about rebuilding, land use, and long-term risk.

Importantly, evaluation reinforced that the system should not be framed as a replacement for expert judgment. Instead, it functions best as a tool that augments human decision-making by organizing information, surfacing trade-offs, and making uncertainty explicit. This aligns with ethical AI principles discussed in class, particularly around accountability, human oversight, and responsible deployment.

6.5 DISCOVERED LIMITATIONS AND SYSTEM CONSTRAINTS

While our system performs well as a decision-support tool, evaluation and feedback revealed several important limitations.

6.5.1 DATA DEPENDENCE AND GENERALIZATION

Model performance is strongly dependent on the quality and coverage of MTBS data. Errors or inconsistencies in burn severity labels propagate into segmentation and downstream risk predictions. Additionally, models were primarily evaluated on California fires, limiting confidence in generalization to regions with different vegetation, climate, or fire regimes. Scaling the system will require additional data and region-specific validation.

6.5.2 ECOLOGICAL AND TEMPORAL SIMPLIFICATIONS

Several ecological factors raised by environmental experts, such as tree spacing, plantation structure, and species composition are not explicitly modeled. These factors can significantly influence fire severity and recovery outcomes, meaning the system may oversimplify complex ecological dynamics. Similarly, the current system relies largely on snapshot-based analysis and does not yet model multi-year recovery trajectories in depth.

6.5.3 UNCERTAINTY AND POLICY CONSTRAINTS

Burn severity labels and risk scores are probabilistic and expert-defined rather than absolute ground truth. This introduces inherent uncertainty that must be carefully communicated to avoid overconfidence. In addition, even well-supported recommendations may not be actionable due to political, legal, or funding constraints. This limits the system's ability to directly influence outcomes and reinforces its role as a decision-support tool rather than an automated decision-maker.

6.5.4 UX MATURITY BASED ON USER FEEDBACK

User feedback highlighted areas where the system needs further refinement. Professional users requested clearer confidence indicators and improved export options. Environmental experts emphasized the need for longer-term scenario views. General users wanted clearer legends, more contextual explanations, and tighter coupling between burn severity and recommended actions. These limitations point to the need for continued UX iteration to support broader adoption.

7 SUMMARY OF MILESTONES

Our original proposal outlined milestones focused on supporting professional post-fire decision-makers, particularly planners and land managers. Over the course of the project, these milestones evolved as evaluation and feedback revealed opportunities to broaden system impact and improve decision support.

Initially, our primary goal was to build a pipeline for burn severity mapping and reburn risk estimation using satellite imagery and historical fire data. This milestone was achieved through iterative development of a segmentation model and a reburn risk classifier, along with evaluation and error analysis to improve reliability.

As the project progressed, we introduced an additional milestone centered on translating predictions into actionable guidance. This led to the development of the Best Next Step optimization module, which was not fully specified in the original proposal. This shift reflected a recognition that users needed prescriptive support, not just descriptive risk maps.

Another major change involved expanding the target user base. While the proposal initially focused on professional users, feedback revealed value for more general audiences such as homebuyers and community members. As a result, we added milestones related to user personas, explainability features, and interface redesign to support multiple levels of expertise.

Some milestones were intentionally deferred or rescaled. Full geographic generalization and long-term temporal modeling were identified as important goals but were not fully achieved due to data and time constraints. These were reframed as future work rather than immediate deliverables.

Overall, milestone changes were driven by evaluation insights rather than technical setbacks. The final system reflects a shift from a narrow analytical tool toward a broader, human-centered decision-support platform aligned with course principles.

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A APPENDIX

You may include other additional sections here.

A.1 EVALUATION

Theme/Feature	Findings	Implications/Quotes
Unified Workflow (One-Stop Shop)	Highly valued for reducing context-switching and synthesis latency; users cited clear efficiency gains	“I like that I don’t have to switch between multiple windows”
Risk/Severity Layer Integration	Enabled fluid sensemaking between current damage and future risk, facilitating rapid hypothesis generation	Improved “operational picture” and decision-making speed
AI/Ask AI Assistant	Transformed data accessibility for lay users, providing plain-language explanations	“I can understand terms like dNBR with the AI”; experts noted: “You could add confidence scores to improve trustworthiness.”
GA-based Recommendations	Actionable output welcomed, but static cost functions and opacity hindered trust and flexibility; lack of user-driven weighting	“The definition of optimal depends on context. I want to adjust the priorities myself.”
Action Plan Granularity	Recommendations sometimes infeasible due to missing constraints (budget, access, land tenure)	Requests for user-entered “real world” constraints
Temporal Data Ambiguity	“Live” labeling misunderstood; confusion over recency and reliability of data	Call for explicit timestamping and “confidence decay” indicators
Scenario-Based Comparison	Historical fire overlays helped users calibrate present decisions, boosting trust and strategic planning capacity	

Table 3: Summary of qualitative findings: key features, participant reactions, and design implications.

A.2 UI SCREENSHOTS

Representative screenshots of the TerraNova workflow are shown below.

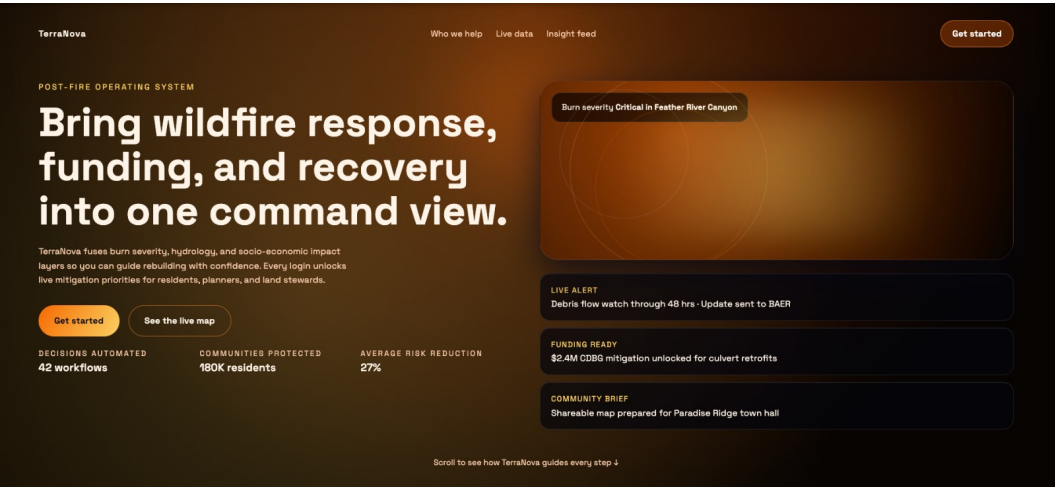


Figure 4: Landing page that orients users to TerraNova (purpose, target user populations, and supporting data sources) and provides a single *Get started* entry point into the analysis workflow.

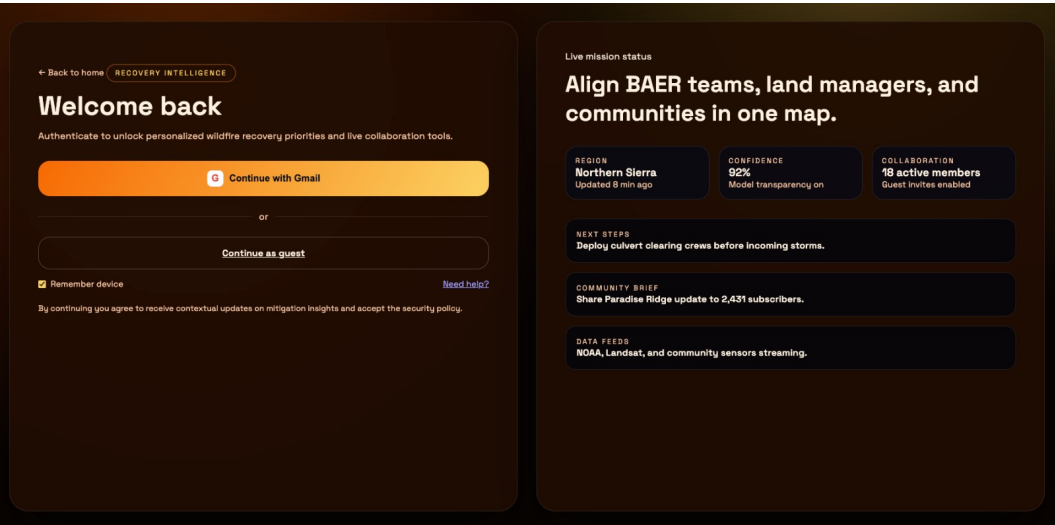


Figure 5: Login and access screen offering Google sign-in or guest access, with brief guidance on intended user groups. This step handles access before exposing domain-specific analysis controls.

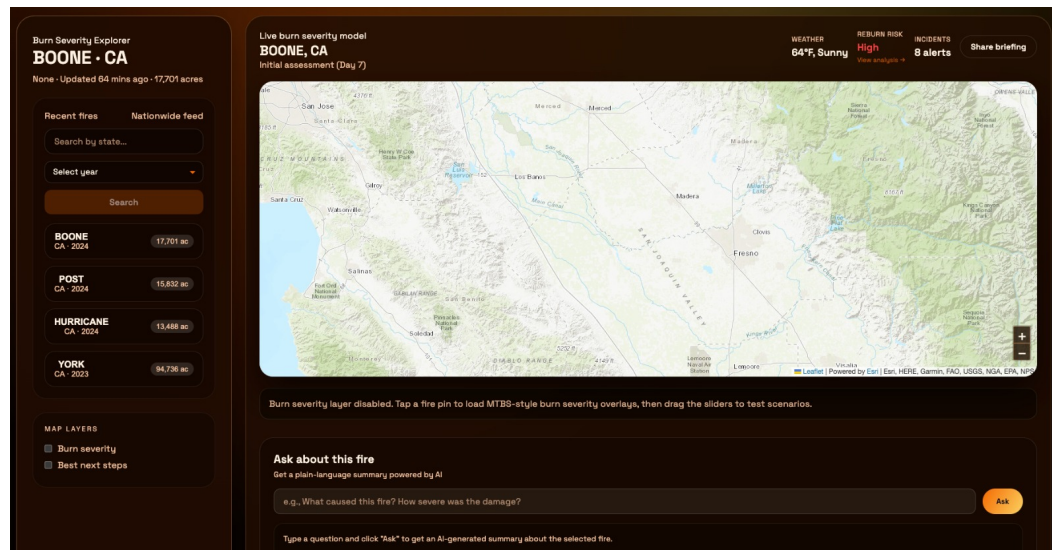


Figure 6: Main analysis screen: an interactive map paired with a control panel for region/year filtering and fire selection, plus entry points to layer toggles, Ask AI, and the analysis dashboard.

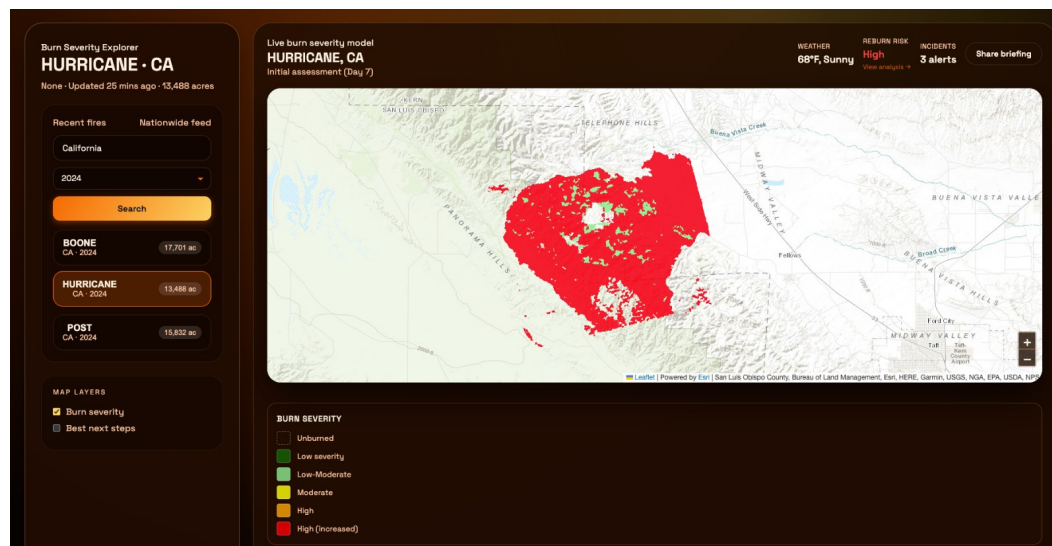


Figure 7: Burn severity layer: U-Net segmentation output rendered as a severity-class overlay for the selected fire, with a legend that makes severity levels directly comparable across the footprint.

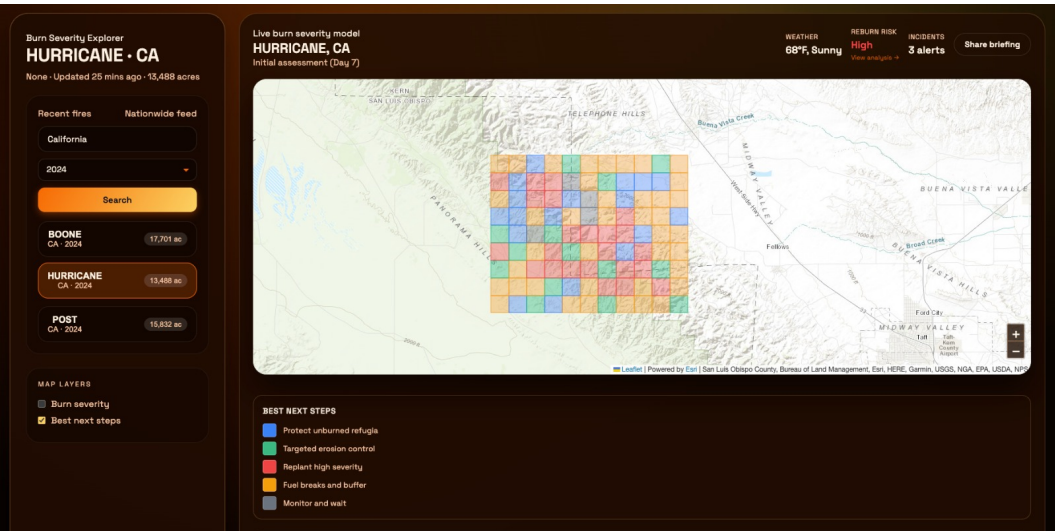


Figure 8: Best next steps layer: optimization output displayed as action categories per grid cell to support prescriptive recovery planning. Users can toggle this layer alongside severity and risk to compare signals before prioritizing interventions.

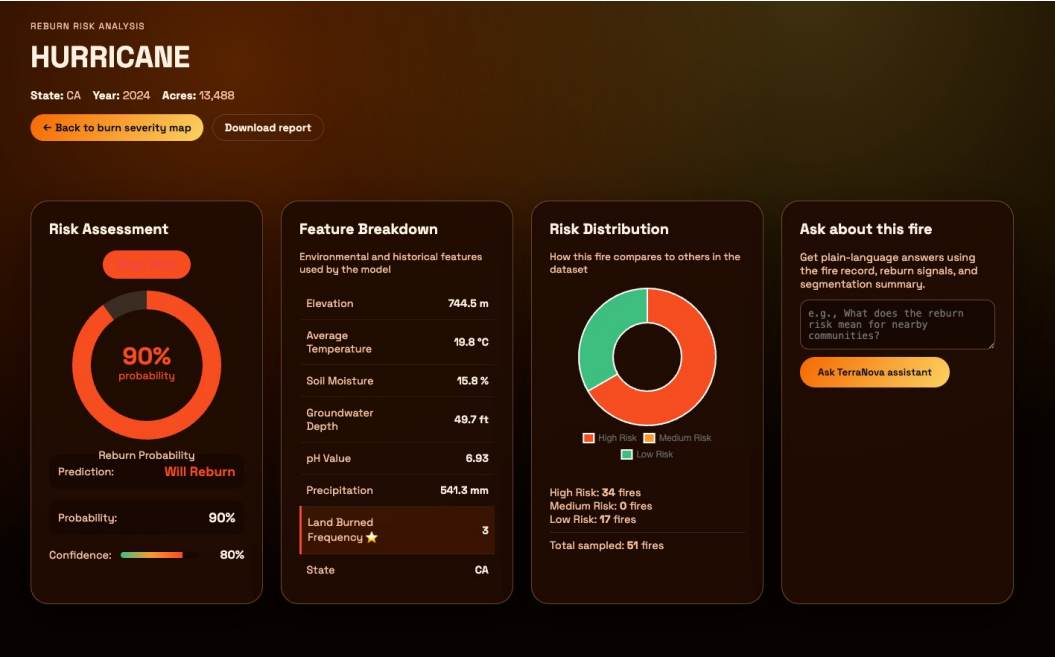


Figure 9: Analysis dashboard: summarizes the predicted reburn probability, provides feature context used by the model, and shows risk distribution relative to other fires. The screen also includes an Ask AI panel for plain-language, context-grounded questions about the selected fire and a report export option to support coordination and handoff.